

SOFTWARE ENGINEERING

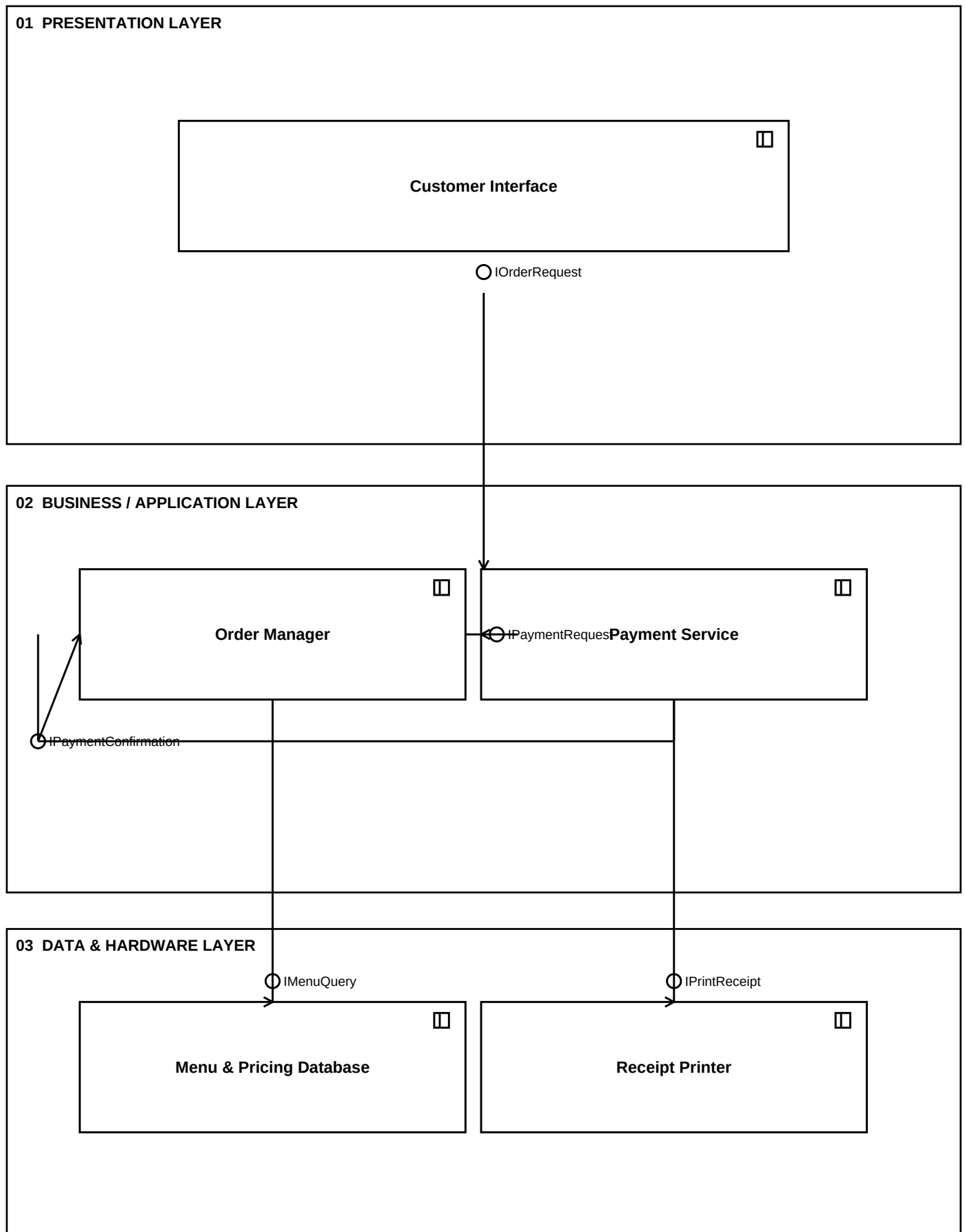
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01 SELF-SERVICE COFFEE KIOSK – COMPONENT DIAGRAM



Provided/required interface points and component connections are shown using UML-style notation.

02 ARCHITECTURE SELECTION

Layered Architecture

We chose the Layered Architecture for the Self-Service Coffee Kiosk System. The first reason for selecting this architecture is clear separation of concerns. The kiosk performs several distinct tasks, including handling touchscreen interactions, managing coffee orders, processing credit-card payments, retrieving menu and pricing data, and printing receipts. By separating these responsibilities into presentation, business/application, and data/hardware responsibilities, the system becomes easier to understand, maintain, test, and extend.

The second reason is that the kiosk requires a structured request-processing flow. A customer's selection first enters through the Customer Interface in the Presentation Layer and is handed to the Order Manager in the Business/Application Layer. The Order Manager retrieves coffee and size prices from the Menu & Pricing Database, calculates the order total, and sends a payment request to the Payment Service. After successful payment, the Order Manager sends the completed order details to the Receipt Printer. This clear flow ensures that each component performs only its own responsibility.

A major security advantage is that payment handling is isolated in the Payment Service rather than being embedded directly in the touchscreen or menu components. This reduces the number of components that need access to sensitive payment operations and allows security controls to be concentrated around the payment interface. Keeping menu and pricing access through a dedicated data component also provides centralized control over stored information.

A key performance benefit is that menu and pricing information can be retrieved efficiently from the dedicated data component, while the Customer Interface remains focused on responsive touchscreen interaction. Separating order processing, payment, data access, and receipt printing also prevents unnecessary work from being performed by the presentation component. Therefore, the Layered Architecture provides a clear, modular, secure, and maintainable solution for the Self-Service Coffee Kiosk System.